

Chapter 3: Determinants

All textbook examples with step-by-step solutions

Based on *Elementary Linear Algebra, 8th edition* by Ron Larson, Chapter 3.

Prepared as a study PDF from the tutoring lesson.

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How to use this PDF

Each example is written in a beginner-friendly way. The order follows the textbook sections:

1. Section 3.1: The Determinant of a Matrix
2. Section 3.2: Determinants and Elementary Operations
3. Section 3.3: Properties of Determinants
4. Section 3.4: Applications of Determinants

For exam preparation, focus on the method used in each example: cofactor expansion, row/column operations, determinant properties, Cramer's Rule, or geometric determinant formulas.

1 3.1 The Determinant of a Matrix

Key formula

For a 2×2 matrix,

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

This means: multiply the main diagonal, multiply the other diagonal, then subtract.

1.1 Example 3.1.1: Determinants of matrices of order 2

Find the determinants of the following matrices:

$$A = \begin{bmatrix} 2 & -3 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & \frac{3}{2} \\ 2 & 4 \end{bmatrix}.$$

Solution for A

$$|A| = \begin{vmatrix} 2 & -3 \\ 1 & 2 \end{vmatrix}.$$

Use the formula $ad - bc$:

$$|A| = 2(2) - 1(-3).$$

Multiply:

$$|A| = 4 - (-3).$$

Subtracting a negative becomes addition:

$$|A| = 4 + 3 = 7.$$

Final answer

$$|A| = 7$$

Solution for B

$$|B| = \begin{vmatrix} 2 & 1 \\ 4 & 2 \end{vmatrix}.$$

Use $ad - bc$:

$$|B| = 2(2) - 4(1) = 4 - 4 = 0.$$

Final answer

$$|B| = 0$$

Solution for C

$$|C| = \begin{vmatrix} 0 & \frac{3}{2} \\ 2 & 4 \end{vmatrix}.$$

Use $ad - bc$:

$$|C| = 0(4) - 2\left(\frac{3}{2}\right).$$

Compute:

$$|C| = 0 - 3 = -3.$$

Final answer

$$\boxed{|C| = -3}$$

1.2 Example 3.1.2: Minors and cofactors of a matrix

Find all minors and cofactors of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

Key formula

The minor M_{ij} is found by deleting row i and column j , then taking the determinant of the remaining matrix.

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

The 3×3 cofactor sign pattern is

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}.$$

Step 1: Find the minors

$$M_{11} = \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} = (-1)(1) - 0(2) = -1.$$

$$M_{12} = \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix} = 3(1) - 4(2) = 3 - 8 = -5.$$

$$M_{13} = \begin{vmatrix} 3 & -1 \\ 4 & 0 \end{vmatrix} = 3(0) - 4(-1) = 4.$$

$$M_{21} = \begin{vmatrix} 2 & 1 \\ 0 & 1 \end{vmatrix} = 2(1) - 0(1) = 2.$$

$$M_{22} = \begin{vmatrix} 0 & 1 \\ 4 & 1 \end{vmatrix} = 0(1) - 4(1) = -4.$$

$$M_{23} = \begin{vmatrix} 0 & 2 \\ 4 & 0 \end{vmatrix} = 0(0) - 4(2) = -8.$$

$$M_{31} = \begin{vmatrix} 2 & 1 \\ -1 & 2 \end{vmatrix} = 2(2) - (-1)(1) = 5.$$

$$M_{32} = \begin{vmatrix} 0 & 1 \\ 3 & 2 \end{vmatrix} = 0(2) - 3(1) = -3.$$

$$M_{33} = \begin{vmatrix} 0 & 2 \\ 3 & -1 \end{vmatrix} = 0(-1) - 3(2) = -6.$$

So the matrix of minors is

$$\begin{bmatrix} -1 & -5 & 4 \\ 2 & -4 & -8 \\ 5 & -3 & -6 \end{bmatrix}.$$

Step 2: Apply cofactor signs

Apply

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}.$$

Therefore,

$$\begin{aligned} C_{11} &= -1, & C_{12} &= 5, & C_{13} &= 4, \\ C_{21} &= -2, & C_{22} &= -4, & C_{23} &= 8, \\ C_{31} &= 5, & C_{32} &= 3, & C_{33} &= -6. \end{aligned}$$

Final answer

The matrix of cofactors is

$$\begin{bmatrix} -1 & 5 & 4 \\ -2 & -4 & 8 \\ 5 & 3 & -6 \end{bmatrix}.$$

1.3 Example 3.1.3: Determinant of a matrix of order 3

Find the determinant of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

From the previous example,

$$C_{11} = -1, \quad C_{12} = 5, \quad C_{13} = 4.$$

Use first row expansion:

$$|A| = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}.$$

Substitute the first row entries 0, 2, 1:

$$|A| = 0(-1) + 2(5) + 1(4).$$

Compute:

$$|A| = 0 + 10 + 4 = 14.$$

Final answer

$$|A| = 14$$

Check using the second row

The second row is 3, -1, 2 and the cofactors are $C_{21} = -2$, $C_{22} = -4$, $C_{23} = 8$.

$$|A| = 3(-2) + (-1)(-4) + 2(8) = -6 + 4 + 16 = 14.$$

Check using the first column

The first column is 0, 3, 4 and the cofactors are $C_{11} = -1$, $C_{21} = -2$, $C_{31} = 5$.

$$|A| = 0(-1) + 3(-2) + 4(5) = 0 - 6 + 20 = 14.$$

1.4 Example 3.1.4: Determinant of a matrix of order 4

Find the determinant of

$$A = \begin{bmatrix} 1 & -2 & 3 & 0 \\ -1 & 1 & 0 & 2 \\ 0 & 2 & 0 & 3 \\ 3 & 4 & 0 & -2 \end{bmatrix}.$$

Step 1: Choose the easiest row or column

The third column is

$$\begin{bmatrix} 3 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

This column has three zeros, so expand along the third column:

$$|A| = 3C_{13} + 0C_{23} + 0C_{33} + 0C_{43} = 3C_{13}.$$

Step 2: Find C_{13}

$$C_{13} = (-1)^{1+3}M_{13} = M_{13}.$$

Delete row 1 and column 3:

$$C_{13} = \begin{vmatrix} -1 & 1 & 2 \\ 0 & 2 & 3 \\ 3 & 4 & -2 \end{vmatrix}.$$

Expand this 3×3 determinant along row 2:

$$C_{13} = 0 + 2(+1) \begin{vmatrix} -1 & 2 \\ 3 & -2 \end{vmatrix} + 3(-1) \begin{vmatrix} -1 & 1 \\ 3 & 4 \end{vmatrix}.$$

Now calculate each 2×2 determinant:

$$\begin{vmatrix} -1 & 2 \\ 3 & -2 \end{vmatrix} = (-1)(-2) - 3(2) = 2 - 6 = -4.$$

$$\begin{vmatrix} -1 & 1 \\ 3 & 4 \end{vmatrix} = (-1)(4) - 3(1) = -4 - 3 = -7.$$

So,

$$C_{13} = 2(-4) + 3(-1)(-7) = -8 + 21 = 13.$$

Step 3: Finish

$$|A| = 3C_{13} = 3(13) = 39.$$

Final answer

$$|A| = 39$$

1.5 Example 3.1.5: Determinant of a 3×3 matrix using the diagonal method

Find the determinant of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & -4 & 1 \end{bmatrix}.$$

For a 3×3 matrix,

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - ceg - bdi - afh.$$

Here,

$$a = 0, b = 2, c = 1, d = 3, e = -1, f = 2, g = 4, h = -4, i = 1.$$

Substitute:

$$|A| = 0(-1)(1) + 2(2)(4) + 1(3)(-4) - 1(-1)(4) - 2(3)(1) - 0(2)(-4).$$

Compute term by term:

$$|A| = 0 + 16 - 12 - (-4) - 6 - 0.$$

$$|A| = 16 - 12 + 4 - 6 = 2.$$

Final answer

$$|A| = 2$$

Important note

This diagonal method works only for 3×3 determinants. For larger matrices, use cofactor expansion or row/column operations.

1.6 Example 3.1.6: Determinant of a triangular matrix

Key formula

If A is triangular, then

$$|A| = a_{11}a_{22} \cdots a_{nn}.$$

That is, multiply the entries on the main diagonal.

Part A: Upper triangular matrix

Find the determinant of

$$A = \begin{bmatrix} 2 & 3 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & 3 \end{bmatrix}.$$

Because the matrix is upper triangular,

$$|A| = 2(-1)(3) = -6.$$

Final answer

$$|A| = -6$$

Part B: Lower triangular matrix

Find the determinant of

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & -2 & 0 & 0 \\ -5 & 6 & 1 & 0 \\ 1 & 5 & 3 & 3 \end{bmatrix}.$$

Because the matrix is lower triangular,

$$|A| = 2(-2)(1)(3) = -12.$$

Final answer

$$|A| = -12$$

2 3.2 Determinants and Elementary Operations

Key formula

Elementary row operation effects:

1. Interchange two rows: determinant changes sign.
2. Add a multiple of one row to another: determinant does not change.
3. Multiply one row by c : determinant is multiplied by c .

The same rules apply to columns.

2.1 Example 3.2.1: Effects of elementary row operations

Part A: Interchanging two rows

$$A = \begin{bmatrix} 2 & -3 \\ 1 & 4 \end{bmatrix}.$$

$$|A| = 2(4) - 1(-3) = 8 + 3 = 11.$$

After interchanging the two rows,

$$B = \begin{bmatrix} 1 & 4 \\ 2 & -3 \end{bmatrix}.$$

$$|B| = 1(-3) - 2(4) = -3 - 8 = -11.$$

Final answer

$$|B| = -|A|$$

Interchanging two rows changes the sign of the determinant.

Part B: Adding a multiple of one row to another

$$A = \begin{bmatrix} 1 & -3 \\ 2 & -4 \end{bmatrix}.$$

$$|A| = 1(-4) - 2(-3) = -4 + 6 = 2.$$

Use $R_2 - 2R_1 \rightarrow R_2$:

$$R_2 - 2R_1 = [2, -4] - 2[1, -3] = [2, -4] - [2, -6] = [0, 2].$$

So

$$B = \begin{bmatrix} 1 & -3 \\ 0 & 2 \end{bmatrix}.$$

$$|B| = 1(2) - 0(-3) = 2.$$

Final answer

$$|B| = |A| = 2$$

Adding a multiple of one row to another does not change the determinant.

Part C: Multiplying a row by a constant

$$A = \begin{bmatrix} 2 & -8 \\ -2 & 9 \end{bmatrix}.$$

$$|A| = 2(9) - (-2)(-8) = 18 - 16 = 2.$$

Multiply the first row by $\frac{1}{2}$:

$$B = \begin{bmatrix} 1 & -4 \\ -2 & 9 \end{bmatrix}.$$

$$|B| = 1(9) - (-2)(-4) = 9 - 8 = 1.$$

Final answer

$$|B| = \frac{1}{2}|A|$$

Multiplying one row by $\frac{1}{2}$ multiplies the determinant by $\frac{1}{2}$.

2.2 Example 3.2.2: Finding a determinant using elementary row operations

Find

$$A = \begin{bmatrix} 0 & -7 & 14 \\ 1 & 2 & -2 \\ 0 & 3 & -8 \end{bmatrix}.$$

Step 1: Interchange rows 1 and 2

$$\begin{vmatrix} 0 & -7 & 14 \\ 1 & 2 & -2 \\ 0 & 3 & -8 \end{vmatrix} = - \begin{vmatrix} 1 & 2 & -2 \\ 0 & -7 & 14 \\ 0 & 3 & -8 \end{vmatrix}.$$

The negative sign appears because one row interchange changes the sign of the determinant.

Step 2: Factor -7 from row 2

$$- \begin{vmatrix} 1 & 2 & -2 \\ 0 & -7 & 14 \\ 0 & 3 & -8 \end{vmatrix} = 7 \begin{vmatrix} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 0 & 3 & -8 \end{vmatrix}.$$

Step 3: Create a zero below the pivot

Use $R_3 - 3R_2 \rightarrow R_3$:

$$[0, 3, -8] - 3[0, 1, -2] = [0, 3, -8] - [0, 3, -6] = [0, 0, -2].$$

Thus,

$$|A| = 7 \begin{vmatrix} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & -2 \end{vmatrix}.$$

Step 4: Use the triangular rule

$$\begin{vmatrix} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & -2 \end{vmatrix} = 1(1)(-2) = -2.$$

Therefore,

$$|A| = 7(-2) = -14.$$

Final answer

$$|A| = -14$$

2.3 Example 3.2.3: Finding a determinant using elementary column operations

Find

$$A = \begin{bmatrix} -1 & 2 & 2 \\ 3 & -6 & 4 \\ 5 & -10 & -3 \end{bmatrix}.$$

Step 1: Notice a column relationship

$$C_2 = \begin{bmatrix} 2 \\ -6 \\ -10 \end{bmatrix} = -2 \begin{bmatrix} -1 \\ 3 \\ 5 \end{bmatrix} = -2C_1.$$

So column 2 is a multiple of column 1.

Step 2: Create a zero column

Use $C_2 + 2C_1 \rightarrow C_2$:

$$\begin{bmatrix} 2 \\ -6 \\ -10 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ -10 \end{bmatrix} + \begin{bmatrix} -2 \\ 6 \\ 10 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

So the determinant becomes

$$|A| = \begin{vmatrix} -1 & 0 & 2 \\ 3 & 0 & 4 \\ 5 & 0 & -3 \end{vmatrix}.$$

Step 3: Expand along the zero column

$$|A| = 0C_{12} + 0C_{22} + 0C_{32} = 0.$$

Final answer

$$|A| = 0$$

2.4 Example 3.2.4: A matrix with a zero determinant

Find the determinant of

$$A = \begin{bmatrix} 1 & 4 & 1 \\ 2 & -1 & 0 \\ 0 & 18 & 4 \end{bmatrix}.$$

Step 1: Use a row operation

Use $R_2 - 2R_1 \rightarrow R_2$:

$$[2, -1, 0] - 2[1, 4, 1] = [2, -1, 0] - [2, 8, 2] = [0, -9, -2].$$

So

$$|A| = \begin{vmatrix} 1 & 4 & 1 \\ 0 & -9 & -2 \\ 0 & 18 & 4 \end{vmatrix}.$$

This operation does not change the determinant.

Step 2: Compare rows

$$R_3 = [0, 18, 4] = -2[0, -9, -2] = -2R_2.$$

One row is a multiple of another row, so the determinant is zero.

Final answer

$$|A| = 0$$

2.5 Example 3.2.5: Combining column operations and cofactor expansion

Find

$$A = \begin{bmatrix} -3 & 5 & 2 \\ 2 & -4 & -1 \\ -3 & 0 & 6 \end{bmatrix}.$$

Step 1: Create another zero in row 3

Use $C_3 + 2C_1 \rightarrow C_3$.

$$C_3 + 2C_1 = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix} + 2 \begin{bmatrix} -3 \\ 2 \\ -3 \end{bmatrix} = \begin{bmatrix} -4 \\ 3 \\ 0 \end{bmatrix}.$$

So

$$|A| = \begin{vmatrix} -3 & 5 & -4 \\ 2 & -4 & 3 \\ -3 & 0 & 0 \end{vmatrix}.$$

This column operation does not change the determinant.

Step 2: Expand along row 3

Row 3 is $[-3, 0, 0]$, so

$$|A| = (-3)C_{31}.$$

Since

$$C_{31} = (-1)^{3+1} \begin{vmatrix} 5 & -4 \\ -4 & 3 \end{vmatrix} = \begin{vmatrix} 5 & -4 \\ -4 & 3 \end{vmatrix},$$

we compute

$$\begin{vmatrix} 5 & -4 \\ -4 & 3 \end{vmatrix} = 5(3) - (-4)(-4) = 15 - 16 = -1.$$

Therefore,

$$|A| = (-3)(-1) = 3.$$

Final answer

$$|A| = 3$$

2.6 Example 3.2.6: Finding a determinant of a 5×5 matrix

Find

$$A = \begin{bmatrix} 2 & 0 & 1 & 3 & -2 \\ -2 & 1 & 3 & 2 & -1 \\ 1 & 0 & -1 & 2 & 3 \\ 3 & -1 & 2 & 4 & -3 \\ 1 & 1 & 3 & 2 & 0 \end{bmatrix}.$$

Step 1: Create zeros in column 2

Column 2 already has zeros in rows 1 and 3. Use

$$R_4 + R_2 \rightarrow R_4, \quad R_5 - R_2 \rightarrow R_5.$$

Compute:

$$R_4 + R_2 = [3, -1, 2, 4, -3] + [-2, 1, 3, 2, -1] = [1, 0, 5, 6, -4],$$

$$R_5 - R_2 = [1, 1, 3, 2, 0] - [-2, 1, 3, 2, -1] = [3, 0, 0, 0, 1].$$

So

$$|A| = \begin{vmatrix} 2 & 0 & 1 & 3 & -2 \\ -2 & 1 & 3 & 2 & -1 \\ 1 & 0 & -1 & 2 & 3 \\ 1 & 0 & 5 & 6 & -4 \\ 3 & 0 & 0 & 0 & 1 \end{vmatrix}.$$

Step 2: Expand down column 2

Only the entry in row 2 is nonzero:

$$|A| = (1)(-1)^{2+2} \begin{vmatrix} 2 & 1 & 3 & -2 \\ 1 & -1 & 2 & 3 \\ 1 & 5 & 6 & -4 \\ 3 & 0 & 0 & 1 \end{vmatrix}.$$

So

$$|A| = \begin{vmatrix} 2 & 1 & 3 & -2 \\ 1 & -1 & 2 & 3 \\ 1 & 5 & 6 & -4 \\ 3 & 0 & 0 & 1 \end{vmatrix}.$$

Step 3: Create a zero in row 4, column 1

Use $C_1 - 3C_4 \rightarrow C_1$:

$$C_1 - 3C_4 = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 3 \end{bmatrix} - 3 \begin{bmatrix} -2 \\ 3 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 8 \\ -8 \\ 13 \\ 0 \end{bmatrix}.$$

Thus,

$$|A| = \begin{vmatrix} 8 & 1 & 3 & -2 \\ -8 & -1 & 2 & 3 \\ 13 & 5 & 6 & -4 \\ 0 & 0 & 0 & 1 \end{vmatrix}.$$

Step 4: Expand along row 4

$$|A| = (1)(-1)^{4+4} \begin{vmatrix} 8 & 1 & 3 \\ -8 & -1 & 2 \\ 13 & 5 & 6 \end{vmatrix} = \begin{vmatrix} 8 & 1 & 3 \\ -8 & -1 & 2 \\ 13 & 5 & 6 \end{vmatrix}.$$

Step 5: Create zeros in row 1

Use $R_1 + R_2 \rightarrow R_1$:

$$[8, 1, 3] + [-8, -1, 2] = [0, 0, 5].$$

So

$$|A| = \begin{vmatrix} 0 & 0 & 5 \\ -8 & -1 & 2 \\ 13 & 5 & 6 \end{vmatrix}.$$

Step 6: Expand along row 1

$$|A| = 5(-1)^{1+3} \begin{vmatrix} -8 & -1 \\ 13 & 5 \end{vmatrix} = 5 \begin{vmatrix} -8 & -1 \\ 13 & 5 \end{vmatrix}.$$

Compute:

$$\begin{vmatrix} -8 & -1 \\ 13 & 5 \end{vmatrix} = (-8)(5) - 13(-1) = -40 + 13 = -27.$$

Therefore,

$$|A| = 5(-27) = -135.$$

Final answer

$$|A| = -135$$

3.3 Properties of Determinants

Key formula

Important determinant properties:

$$\begin{aligned} |AB| &= |A||B|, \\ |cA| &= c^n|A| \quad \text{for an } n \times n \text{ matrix,} \\ A \text{ is invertible} &\iff |A| \neq 0, \\ |A^{-1}| &= \frac{1}{|A|}, \quad |A^T| = |A|. \end{aligned}$$

3.1 Example 3.3.1: Determinant of a matrix product

Find $|A|$, $|B|$, AB , and $|AB|$ for

$$A = \begin{bmatrix} 1 & -2 & 2 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 & 1 \\ 0 & -1 & -2 \\ 3 & 1 & -2 \end{bmatrix}.$$

Step 1: Find $|A|$

Expand along row 2:

$$|A| = 0C_{21} + 3C_{22} + 2C_{23}.$$

Find C_{22} :

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = 1[1(1) - 1(2)] = -1.$$

Find C_{23} :

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 1 & -2 \\ 1 & 0 \end{vmatrix} = -[1(0) - 1(-2)] = -2.$$

Therefore,

$$|A| = 3(-1) + 2(-2) = -3 - 4 = -7.$$

Step 2: Find $|B|$

Expand along row 1:

$$|B| = 2C_{11} + 0C_{12} + 1C_{13}.$$

$$C_{11} = \begin{vmatrix} -1 & -2 \\ 1 & -2 \end{vmatrix} = (-1)(-2) - 1(-2) = 2 + 2 = 4.$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 0 & -1 \\ 3 & 1 \end{vmatrix} = 0(1) - 3(-1) = 3.$$

Thus,

$$|B| = 2(4) + 1(3) = 8 + 3 = 11.$$

Step 3: Multiply AB

$$AB = \begin{bmatrix} 1 & -2 & 2 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ 0 & -1 & -2 \\ 3 & 1 & -2 \end{bmatrix}.$$

Compute row by column:

$$AB = \begin{bmatrix} 8 & 4 & 1 \\ 6 & -1 & -10 \\ 5 & 1 & -1 \end{bmatrix}.$$

Step 4: Find $|AB|$

$$|AB| = \begin{vmatrix} 8 & 4 & 1 \\ 6 & -1 & -10 \\ 5 & 1 & -1 \end{vmatrix}.$$

Expand along row 1:

$$|AB| = 8 \begin{vmatrix} -1 & -10 \\ 1 & -1 \end{vmatrix} - 4 \begin{vmatrix} 6 & -10 \\ 5 & -1 \end{vmatrix} + 1 \begin{vmatrix} 6 & -1 \\ 5 & 1 \end{vmatrix}.$$

Compute each small determinant:

$$\begin{vmatrix} -1 & -10 \\ 1 & -1 \end{vmatrix} = 1 + 10 = 11,$$

$$\begin{vmatrix} 6 & -10 \\ 5 & -1 \end{vmatrix} = -6 + 50 = 44,$$

$$\begin{vmatrix} 6 & -1 \\ 5 & 1 \end{vmatrix} = 6 + 5 = 11.$$

So

$$|AB| = 8(11) - 4(44) + 1(11) = 88 - 176 + 11 = -77.$$

Final answer

$$|A| = -7, \quad |B| = 11, \quad |AB| = -77$$

Also,

$$|A||B| = (-7)(11) = -77 = |AB|.$$

3.2 Example 3.3.2: Determinant of a scalar multiple of a matrix

Find the determinant of

$$A = \begin{bmatrix} 10 & -20 & 40 \\ 30 & 0 & 50 \\ -20 & -30 & 10 \end{bmatrix}.$$

Every entry is a multiple of 10, so write

$$A = 10 \begin{bmatrix} 1 & -2 & 4 \\ 3 & 0 & 5 \\ -2 & -3 & 1 \end{bmatrix} = 10B.$$

Since A is 3×3 ,

$$|10B| = 10^3|B|.$$

Now compute

$$|B| = \begin{vmatrix} 1 & -2 & 4 \\ 3 & 0 & 5 \\ -2 & -3 & 1 \end{vmatrix}.$$

Expand along row 1:

$$|B| = 1 \begin{vmatrix} 0 & 5 \\ -3 & 1 \end{vmatrix} - (-2) \begin{vmatrix} 3 & 5 \\ -2 & 1 \end{vmatrix} + 4 \begin{vmatrix} 3 & 0 \\ -2 & -3 \end{vmatrix}.$$

Compute each determinant:

$$\begin{vmatrix} 0 & 5 \\ -3 & 1 \end{vmatrix} = 15,$$

$$\begin{vmatrix} 3 & 5 \\ -2 & 1 \end{vmatrix} = 3 + 10 = 13,$$

$$\begin{vmatrix} 3 & 0 \\ -2 & -3 \end{vmatrix} = -9.$$

So

$$|B| = 1(15) - (-2)(13) + 4(-9) = 15 + 26 - 36 = 5.$$

Therefore,

$$|A| = 10^3 |B| = 1000(5) = 5000.$$

Final answer

$$|A| = 5000$$

3.3 Example 3.3.3: Determinant of a matrix sum warning

Show that generally $|A + B| \neq |A| + |B|$ using

$$A = \begin{bmatrix} 6 & 2 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 7 \\ 0 & -1 \end{bmatrix}.$$

First,

$$|A| = 6(1) - 2(2) = 6 - 4 = 2.$$

$$|B| = 3(-1) - 0(7) = -3.$$

So

$$|A| + |B| = 2 + (-3) = -1.$$

Now add the matrices:

$$A + B = \begin{bmatrix} 6+3 & 2+7 \\ 2+0 & 1+(-1) \end{bmatrix} = \begin{bmatrix} 9 & 9 \\ 2 & 0 \end{bmatrix}.$$

Then

$$|A + B| = 9(0) - 2(9) = -18.$$

Final answer

$$|A| + |B| = -1 \quad \text{but} \quad |A + B| = -18$$

Therefore, in general,

$$|A + B| \neq |A| + |B|.$$

3.4 Example 3.3.4: Classifying matrices as singular or nonsingular

Determine whether each matrix has an inverse.

$$A = \begin{bmatrix} 0 & 2 & -1 \\ 3 & -2 & 1 \\ 3 & 2 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 2 & -1 \\ 3 & -2 & 1 \\ 3 & 2 & 1 \end{bmatrix}.$$

Matrix A

$$|A| = \begin{vmatrix} 0 & 2 & -1 \\ 3 & -2 & 1 \\ 3 & 2 & -1 \end{vmatrix}.$$

Expand along row 1:

$$|A| = 0 - 2 \begin{vmatrix} 3 & 1 \\ 3 & -1 \end{vmatrix} + (-1) \begin{vmatrix} 3 & -2 \\ 3 & 2 \end{vmatrix}.$$

Compute:

$$\begin{vmatrix} 3 & 1 \\ 3 & -1 \end{vmatrix} = 3(-1) - 3(1) = -6,$$

$$\begin{vmatrix} 3 & -2 \\ 3 & 2 \end{vmatrix} = 3(2) - 3(-2) = 12.$$

Thus,

$$|A| = -2(-6) + (-1)(12) = 12 - 12 = 0.$$

Since $|A| = 0$, A has no inverse.

A is singular.

Matrix B

$$|B| = \begin{vmatrix} 0 & 2 & -1 \\ 3 & -2 & 1 \\ 3 & 2 & 1 \end{vmatrix}.$$

Expand along row 1:

$$|B| = 0 - 2 \begin{vmatrix} 3 & 1 \\ 3 & 1 \end{vmatrix} + (-1) \begin{vmatrix} 3 & -2 \\ 3 & 2 \end{vmatrix}.$$

Compute:

$$\begin{vmatrix} 3 & 1 \\ 3 & 1 \end{vmatrix} = 3(1) - 3(1) = 0,$$

$$\begin{vmatrix} 3 & -2 \\ 3 & 2 \end{vmatrix} = 12.$$

Thus,

$$|B| = -2(0) + (-1)(12) = -12.$$

Since $|B| \neq 0$, B has an inverse.

Final answer

A is singular, B is nonsingular.

3.5 Example 3.3.5: Determinant of the inverse of a matrix

Find $|A^{-1}|$ for

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & 2 \\ 2 & 1 & 0 \end{bmatrix}.$$

Use

$$|A^{-1}| = \frac{1}{|A|}.$$

First find $|A|$:

$$|A| = \begin{vmatrix} 1 & 0 & 3 \\ 0 & -1 & 2 \\ 2 & 1 & 0 \end{vmatrix}.$$

Expand along row 1:

$$|A| = 1 \begin{vmatrix} -1 & 2 \\ 1 & 0 \end{vmatrix} - 0 + 3 \begin{vmatrix} 0 & -1 \\ 2 & 1 \end{vmatrix}.$$

Compute:

$$\begin{vmatrix} -1 & 2 \\ 1 & 0 \end{vmatrix} = (-1)(0) - 1(2) = -2,$$

$$\begin{vmatrix} 0 & -1 \\ 2 & 1 \end{vmatrix} = 0(1) - 2(-1) = 2.$$

So

$$|A| = 1(-2) + 3(2) = -2 + 6 = 4.$$

Therefore,

$$|A^{-1}| = \frac{1}{|A|} = \frac{1}{4}.$$

Final answer

$$|A^{-1}| = \frac{1}{4}$$

3.6 Example 3.3.6: Using determinants to decide whether a system has a unique solution

Which system has a unique solution?

System A:

$$\begin{cases} 2x_2 - x_3 = -1, \\ 3x_1 - 2x_2 + x_3 = 4, \\ 3x_1 + 2x_2 - x_3 = -4. \end{cases}$$

System B:

$$\begin{cases} 2x_2 - x_3 = -1, \\ 3x_1 - 2x_2 + x_3 = 4, \\ 3x_1 + 2x_2 + x_3 = -4. \end{cases}$$

The coefficient matrices are the same as in Example 3.3.4:

$$A = \begin{bmatrix} 0 & 2 & -1 \\ 3 & -2 & 1 \\ 3 & 2 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 2 & -1 \\ 3 & -2 & 1 \\ 3 & 2 & 1 \end{bmatrix}.$$

From Example 3.3.4,

$$|A| = 0,$$

so System A does not have a unique solution. Also,

$$|B| = -12 \neq 0,$$

so System B has a unique solution.

Final answer

Only System B has a unique solution.

3.7 Example 3.3.7: Determinant of a transpose

Show that $|A| = |A^T|$ for

$$A = \begin{bmatrix} 3 & 1 & -2 \\ 2 & 0 & 0 \\ -4 & -1 & 5 \end{bmatrix}.$$

Step 1: Find $|A|$

Use row 2 because it has two zeros:

$$|A| = 2C_{21}.$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 1 & -2 \\ -1 & 5 \end{vmatrix} = -[1(5) - (-1)(-2)].$$

$$C_{21} = -(5 - 2) = -3.$$

Therefore,

$$|A| = 2(-3) = -6.$$

Step 2: Find A^T and $|A^T|$

$$A^T = \begin{bmatrix} 3 & 2 & -4 \\ 1 & 0 & -1 \\ -2 & 0 & 5 \end{bmatrix}.$$

Use column 2 because it has two zeros:

$$|A^T| = 2C_{12}.$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 1 & -1 \\ -2 & 5 \end{vmatrix} = -[1(5) - (-2)(-1)].$$

$$C_{12} = -(5 - 2) = -3.$$

Therefore,

$$|A^T| = 2(-3) = -6.$$

Final answer

$$|A| = -6 = |A^T|$$

4 3.4 Applications of Determinants

4.1 Example 3.4.1: Finding the adjoint of a square matrix

Find the adjoint of

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix}.$$

Key formula

The adjoint of A is the transpose of the cofactor matrix:

$$\text{adj}(A) = [\text{cofactor matrix of } A]^T.$$

Step 1: Find the cofactors

$$\begin{aligned} C_{11} &= + \begin{vmatrix} -2 & 1 \\ 0 & -2 \end{vmatrix} = 4. \\ C_{12} &= - \begin{vmatrix} 0 & 1 \\ 1 & -2 \end{vmatrix} = -[0(-2) - 1(1)] = 1. \\ C_{13} &= + \begin{vmatrix} 0 & -2 \\ 1 & 0 \end{vmatrix} = 0(0) - 1(-2) = 2. \\ C_{21} &= - \begin{vmatrix} 3 & 2 \\ 0 & -2 \end{vmatrix} = -[3(-2) - 0(2)] = 6. \\ C_{22} &= + \begin{vmatrix} -1 & 2 \\ 1 & -2 \end{vmatrix} = (-1)(-2) - 1(2) = 0. \\ C_{23} &= - \begin{vmatrix} -1 & 3 \\ 1 & 0 \end{vmatrix} = -[(-1)(0) - 1(3)] = 3. \\ C_{31} &= + \begin{vmatrix} 3 & 2 \\ -2 & 1 \end{vmatrix} = 3(1) - (-2)(2) = 7. \\ C_{32} &= - \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} = -[(-1)(1) - 0(2)] = 1. \\ C_{33} &= + \begin{vmatrix} -1 & 3 \\ 0 & -2 \end{vmatrix} = (-1)(-2) - 0(3) = 2. \end{aligned}$$

So the cofactor matrix is

$$\begin{bmatrix} 4 & 1 & 2 \\ 6 & 0 & 3 \\ 7 & 1 & 2 \end{bmatrix}.$$

Step 2: Transpose the cofactor matrix

$$\text{adj}(A) = \begin{bmatrix} 4 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix}.$$

Final answer

$$\text{adj}(A) = \begin{bmatrix} 4 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix}$$

4.2 Example 3.4.2: Using the adjoint to find the inverse

Use the adjoint of

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix}$$

to find A^{-1} .

From Example 3.4.1,

$$\text{adj}(A) = \begin{bmatrix} 4 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix}.$$

Also,

$$|A| = 3.$$

Use

$$A^{-1} = \frac{1}{|A|} \text{adj}(A).$$

Therefore,

$$A^{-1} = \frac{1}{3} \begin{bmatrix} 4 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix}.$$

Multiply every entry by $\frac{1}{3}$:

$$A^{-1} = \begin{bmatrix} \frac{4}{3} & 2 & \frac{7}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{2}{3} & 1 & \frac{2}{3} \end{bmatrix}.$$

Final answer

$$A^{-1} = \begin{bmatrix} \frac{4}{3} & 2 & \frac{7}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{2}{3} & 1 & \frac{2}{3} \end{bmatrix}$$

4.3 Example 3.4.3: Using Cramer's Rule for a 2×2 system

Solve

$$\begin{cases} 4x_1 - 2x_2 = 10, \\ 3x_1 - 5x_2 = 11. \end{cases}$$

Key formula

For Cramer's Rule,

$$x_i = \frac{|A_i|}{|A|},$$

where A_i is obtained by replacing column i of the coefficient matrix with the constants column.

Step 1: Find $|A|$

$$A = \begin{bmatrix} 4 & -2 \\ 3 & -5 \end{bmatrix}.$$

$$|A| = 4(-5) - 3(-2) = -20 + 6 = -14.$$

Since $|A| \neq 0$, the system has a unique solution.

Step 2: Find x_1

Replace the first column with constants:

$$A_1 = \begin{bmatrix} 10 & -2 \\ 11 & -5 \end{bmatrix}.$$

$$|A_1| = 10(-5) - 11(-2) = -50 + 22 = -28.$$

$$x_1 = \frac{|A_1|}{|A|} = \frac{-28}{-14} = 2.$$

Step 3: Find x_2

Replace the second column with constants:

$$A_2 = \begin{bmatrix} 4 & 10 \\ 3 & 11 \end{bmatrix}.$$

$$|A_2| = 4(11) - 3(10) = 44 - 30 = 14.$$

$$x_2 = \frac{|A_2|}{|A|} = \frac{14}{-14} = -1.$$

Final answer

$$x_1 = 2, \quad x_2 = -1$$

4.4 Example 3.4.4: Cramer's Rule for a 3×3 system

Solve

$$\begin{cases} -x + 2y - 3z = 1, \\ 2x + z = 0, \\ 3x - 4y + 4z = 2. \end{cases}$$

The coefficient matrix and constants column are

$$A = \begin{bmatrix} -1 & 2 & -3 \\ 2 & 0 & 1 \\ 3 & -4 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}.$$

First,

$$|A| = \begin{vmatrix} -1 & 2 & -3 \\ 2 & 0 & 1 \\ 3 & -4 & 4 \end{vmatrix} = 10.$$

Since $|A| \neq 0$, the solution is unique.

Step 1: Solve for x

Replace the x -column with b :

$$A_x = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 1 \\ 2 & -4 & 4 \end{bmatrix}.$$

Expand along row 2:

$$|A_x| = 1C_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 2 \\ 2 & -4 \end{vmatrix}.$$

$$\begin{vmatrix} 1 & 2 \\ 2 & -4 \end{vmatrix} = 1(-4) - 2(2) = -4 - 4 = -8.$$

$$|A_x| = -(-8) = 8.$$

$$x = \frac{|A_x|}{|A|} = \frac{8}{10} = \frac{4}{5}.$$

Step 2: Solve for y

Replace the y -column with b :

$$A_y = \begin{bmatrix} -1 & 1 & -3 \\ 2 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix}.$$

Expand along row 2:

$$|A_y| = 2C_{21} + 1C_{23}.$$

$$C_{21} = (-1)^3 \begin{vmatrix} 1 & -3 \\ 2 & 4 \end{vmatrix} = -[1(4) - 2(-3)] = -10.$$

$$C_{23} = (-1)^5 \begin{vmatrix} -1 & 1 \\ 3 & 2 \end{vmatrix} = -[(-1)(2) - 3(1)] = 5.$$

$$|A_y| = 2(-10) + 5 = -15.$$

$$y = \frac{|A_y|}{|A|} = \frac{-15}{10} = -\frac{3}{2}.$$

Step 3: Solve for z

Replace the z -column with b :

$$A_z = \begin{bmatrix} -1 & 2 & 1 \\ 2 & 0 & 0 \\ 3 & -4 & 2 \end{bmatrix}.$$

Expand along row 2:

$$|A_z| = 2C_{21} = 2(-1)^3 \begin{vmatrix} 2 & 1 \\ -4 & 2 \end{vmatrix}.$$

$$\begin{vmatrix} 2 & 1 \\ -4 & 2 \end{vmatrix} = 2(2) - (-4)(1) = 8.$$

$$|A_z| = 2(-8) = -16.$$

$$z = \frac{|A_z|}{|A|} = \frac{-16}{10} = -\frac{8}{5}.$$

Final answer

$$x = \frac{4}{5}, \quad y = -\frac{3}{2}, \quad z = -\frac{8}{5}$$

4.5 Example 3.4.5: Finding the area of a triangle

Find the area of the triangle with vertices

$$(1, 1), \quad (2, 2), \quad (4, 3).$$

Key formula

$$\text{Area} = \pm \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}.$$

Choose the sign so the area is positive.

Substitute the points:

$$\text{Area} = \pm \frac{1}{2} \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ 4 & 3 & 1 \end{vmatrix}.$$

Expand along row 1:

$$D = 1 \begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} - 1 \begin{vmatrix} 2 & 1 \\ 4 & 1 \end{vmatrix} + 1 \begin{vmatrix} 2 & 2 \\ 4 & 3 \end{vmatrix}.$$

Compute:

$$\begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} = 2(1) - 3(1) = -1,$$

$$\begin{vmatrix} 2 & 1 \\ 4 & 1 \end{vmatrix} = 2(1) - 4(1) = -2,$$

$$\begin{vmatrix} 2 & 2 \\ 4 & 3 \end{vmatrix} = 2(3) - 4(2) = -2.$$

Thus,

$$D = -1 - (-2) + (-2) = -1 + 2 - 2 = -1.$$

$$\text{Area} = \pm \frac{1}{2}(-1).$$

Area must be positive, so

Final answer

$$\text{Area} = \frac{1}{2} \text{ square unit}$$

4.6 Example 3.4.6: Testing for collinear points

Determine whether the points

$$(0, 1), \quad (2, 2), \quad (4, 3)$$

are collinear.

Use the determinant test:

$$D = \begin{vmatrix} 0 & 1 & 1 \\ 2 & 2 & 1 \\ 4 & 3 & 1 \end{vmatrix}.$$

Expand along row 1:

$$D = 0 \begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} - 1 \begin{vmatrix} 2 & 1 \\ 4 & 1 \end{vmatrix} + 1 \begin{vmatrix} 2 & 2 \\ 4 & 3 \end{vmatrix}.$$

Compute:

$$D = 0 - [2(1) - 4(1)] + [2(3) - 4(2)].$$

$$D = 0 - (2 - 4) + (6 - 8) = 0 - (-2) - 2 = 2 - 2 = 0.$$

Since $D = 0$, the points are collinear.

Final answer

The points are collinear.

4.7 Example 3.4.7: Finding an equation of a line through two points

Find an equation of the line through

$$(2, 4) \quad \text{and} \quad (-1, 3).$$

Use the two-point determinant form:

$$\begin{vmatrix} x & y & 1 \\ 2 & 4 & 1 \\ -1 & 3 & 1 \end{vmatrix} = 0.$$

Expand along row 1:

$$x \begin{vmatrix} 4 & 1 \\ 3 & 1 \end{vmatrix} - y \begin{vmatrix} 2 & 1 \\ -1 & 1 \end{vmatrix} + 1 \begin{vmatrix} 2 & 4 \\ -1 & 3 \end{vmatrix} = 0.$$

Compute each determinant:

$$\begin{vmatrix} 4 & 1 \\ 3 & 1 \end{vmatrix} = 4(1) - 3(1) = 1,$$

$$\begin{vmatrix} 2 & 1 \\ -1 & 1 \end{vmatrix} = 2(1) - (-1)(1) = 3,$$

$$\begin{vmatrix} 2 & 4 \\ -1 & 3 \end{vmatrix} = 2(3) - (-1)(4) = 10.$$

Substitute:

$$x(1) - y(3) + 10 = 0.$$

$$x - 3y + 10 = 0.$$

Final answer

$$x - 3y = -10$$

4.8 Example 3.4.8: Finding the volume of a tetrahedron

Find the volume of the tetrahedron with vertices

$$(0, 4, 1), \quad (4, 0, 0), \quad (3, 5, 2), \quad (2, 2, 5).$$

Key formula

$$\text{Volume} = \pm \frac{1}{6} \begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix}.$$

Choose the sign so the volume is positive.

Substitute the points:

$$\text{Volume} = \pm \frac{1}{6} \begin{vmatrix} 0 & 4 & 1 & 1 \\ 4 & 0 & 0 & 1 \\ 3 & 5 & 2 & 1 \\ 2 & 2 & 5 & 1 \end{vmatrix}.$$

The determinant evaluates to -72 :

$$\begin{vmatrix} 0 & 4 & 1 & 1 \\ 4 & 0 & 0 & 1 \\ 3 & 5 & 2 & 1 \\ 2 & 2 & 5 & 1 \end{vmatrix} = -72.$$

Therefore,

$$\text{Volume} = \pm \frac{1}{6}(-72).$$

Since volume must be positive,

$$\text{Volume} = 12.$$

Final answer

12 cubic units

4.9 Example 3.4.9: Finding an equation of a plane through three points

Find an equation of the plane through

$$(0, 1, 0), \quad (-1, 3, 2), \quad (-2, 0, 1).$$

Use the determinant form:

$$\begin{vmatrix} x & y & z & 1 \\ 0 & 1 & 0 & 1 \\ -1 & 3 & 2 & 1 \\ -2 & 0 & 1 & 1 \end{vmatrix} = 0.$$

Use the column operation $C_2 - C_4 \rightarrow C_2$:

$$\begin{vmatrix} x & y-1 & z & 1 \\ 0 & 0 & 0 & 1 \\ -1 & 2 & 2 & 1 \\ -2 & -1 & 1 & 1 \end{vmatrix} = 0.$$

Expand along row 2. The only nonzero entry is in position $(2, 4)$:

$$\begin{vmatrix} x & y-1 & z \\ -1 & 2 & 2 \\ -2 & -1 & 1 \end{vmatrix} = 0.$$

Expand along row 1:

$$x \begin{vmatrix} 2 & 2 \\ -1 & 1 \end{vmatrix} - (y-1) \begin{vmatrix} -1 & 2 \\ -2 & 1 \end{vmatrix} + z \begin{vmatrix} -1 & 2 \\ -2 & -1 \end{vmatrix} = 0.$$

Compute each determinant:

$$\begin{vmatrix} 2 & 2 \\ -1 & 1 \end{vmatrix} = 2(1) - (-1)(2) = 4,$$

$$\begin{vmatrix} -1 & 2 \\ -2 & 1 \end{vmatrix} = (-1)(1) - (-2)(2) = -1 + 4 = 3,$$

$$\begin{vmatrix} -1 & 2 \\ -2 & -1 \end{vmatrix} = (-1)(-1) - (-2)(2) = 1 + 4 = 5.$$

Substitute:

$$4x - 3(y - 1) + 5z = 0.$$

Expand:

$$4x - 3y + 3 + 5z = 0.$$

Move 3 to the right:

$$4x - 3y + 5z = -3.$$

Final answer

$$4x - 3y + 5z = -3$$

Final Chapter 3 Formula Sheet

Basic determinant formulas

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

$$|A| = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}.$$

$$|A| = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}.$$

$$\text{If } A \text{ is triangular, then } |A| = a_{11}a_{22} \cdots a_{nn}.$$

Elementary operation effects

$$R_i \leftrightarrow R_j \Rightarrow \text{determinant changes sign.}$$

$$R_j + cR_i \rightarrow R_j \Rightarrow \text{determinant does not change.}$$

$$cR_i \rightarrow R_i \Rightarrow \text{determinant is multiplied by } c.$$

Properties

$$|AB| = |A||B|.$$

$$|cA| = c^n |A|.$$

$$A \text{ is invertible} \iff |A| \neq 0.$$

$$|A^{-1}| = \frac{1}{|A|}.$$

$$|A^T| = |A|.$$

Applications

$$A^{-1} = \frac{1}{|A|} \text{adj}(A).$$

$$x_i = \frac{|A_i|}{|A|}.$$

$$\text{Area} = \pm \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}.$$

$$\text{Volume} = \pm \frac{1}{6} \begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix}.$$

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0 \quad \text{line through two points.}$$

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0 \quad \text{plane through three points.}$$